

The development of broadband signalling platforms

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Mao Zedong once said: "All genuine knowledge originates in direct experience." This paper describes the experiences gained through a range of development projects, all concerned with the design, development, demonstration and maintenance of real broadband signalling systems. The authors have participated in the development of a variety of systems offering broadband capabilities, all of which support some form of broadband signalling between the service user and the service provider. The process of designing and building such systems can provide invaluable insights into issues such as development methods, signalling performance, programming interface definitions, etc, none of which are apparent from purely paper-based studies or through the standards specification process.

This 'coal-face' experience covers both collaborative as well as internal BT projects, with each system being subject to a wide variety of service requirements. In addition, the period of work covered by the developments described in the paper encompasses broadband signalling systems that on the one hand predate the published ITU-T standards on broadband ISDN systems and on the other include a comprehensive testbed supporting an interworking set of broadband signalling interfaces based on the appropriate signalling standards defined by both the ATM Forum as well as the ITU-T.

1. Introduction

This paper describes a range of development projects, all of which have resulted in the construction of a broadband ISDN platform, and extending as far back as the early 1990s, up until the present day. In (approximately) historical order these are: the RACE I BUNI demonstrator project, the ADP1 Signalling Demonstrator, the RACE II TRIBUNE project, and the FINDS platform. Each of these very different developments is described in further detail below; in particular, the important lessons learnt from the development are spelt out in each case.

2. The BUNI demonstrator

2.1 Introduction

The European Community funded a number of projects in the early 1990s that shared a common aim — the exploration of broadband communications. Under the umbrella of the RACE I (research and development into advanced communication technology in Europe) programme, many companies and institutions were brought together to work on the emerging broadband technologies. The RACE I programme had the further aim of bringing together the various pieces of technology being developed in the various projects to form a comprehensive demonstration of the emerging broadband technology. The broadband user/network interface demonstrator project, commonly known as BUNI demonstrator, was one of these projects that integrated various pieces of equipment

developed in the different RACE I projects with a signalling and control architecture to form a comprehensive demonstration of broadband technology.

The BUNI demonstration network included equipment ranging from multiservice terminals, HDTV encoders/decoders and displays, customer premises equipment (CPE), customer access systems, ATM switches and the all important signalling and control.

At the height of the project upwards of 18 European companies worked on the project; some of these companies also represented other consortiums and hence the total number of companies and contacts associated with the project was large. The composition of companies ranged from the large telecommunications suppliers, through network operators to small start-up companies, so a good mix of cultures, working ethics and styles were represented — it made for an interesting time, as BT was the prime contractor.

2.2 Demonstration environment

The BUNI demonstrator was housed in specially designed accommodation at the laboratories of PTT-Research in Leidschendam, The Netherlands. The purpose-built demonstration facility was designed to display the terminal equipment in three separate, though linked, areas, each reflecting the architectural style of the three distinct 'markets' being addressed. Thus the domestic equipment

was displayed in a 'mock-up' of a home environment, e.g. a lounge, while the business equipment was shown in an office environment, and the broadcast in a studio environment. There were other separate areas to house the network equipment and a reception/lecture area.

The services that could be demonstrated were — conventional and HDTV distribution, multiservice terminals which had a range of narrowband ISDN (N-ISDN) services accessible via the broadband network, and a range of domestic services that included video surveillance, TV, VTR, telephony, etc. The equipment, both network and terminal, to support this range of services was a combination of equipment developed within the BUNI demonstrator project and from other RACE projects. The one common element to all the equipment, whether developed as part of the BUNI demonstrator project or from other RACE projects, was that they all had to conform to the BUNI signalling specification or its related control architecture to enable full interoperability. Figure 1 shows the position of the broadband user/network interface in relationship to the customer premises equipment (TV encoders, terminals, etc) and the network equipment (connection admission control (CAC), switches, N-ISDN gateway, and signalling and control network).

2.3 Signalling specification

The major interface in any broadband network is the broadband user/network interface; this is the point at which the customer premises equipment meets the network operators' equipment. This specification was therefore the major interface within the BUNI demonstrator project and, as such, considerable effort went into the development of the specification for this interface. BT led this activity and was supported by a number of other partners in the project consortium.

The specification followed the cubic approach as depicted in Fig 2. At the time the specification was produced the international standards activity was still ongoing especially in the access signalling area. The approach adopted when producing the specification was that it should be high in functionality and contain enough detail to be unambiguous and hence allow interoperability between the various implementors' equipment. The lower layers of the specification, i.e. physical, ATM layer and ATM adaptation layer, were fairly well specified, in draft form, by the likes of International Telecommunications Union (ITU), European Telecommunications Standards Institute (ETSI), etc. The major task of the BUNI demonstrator partners was to pick which options of the

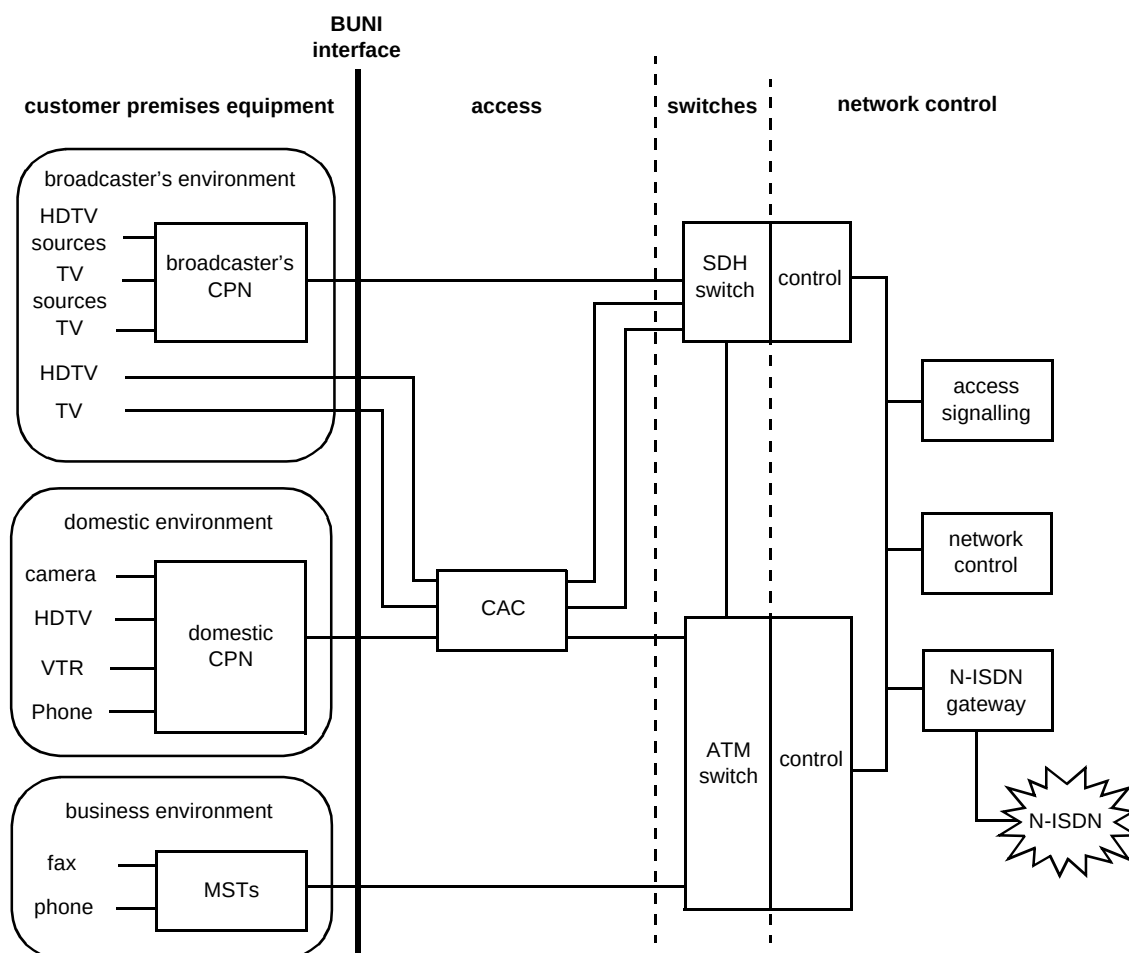
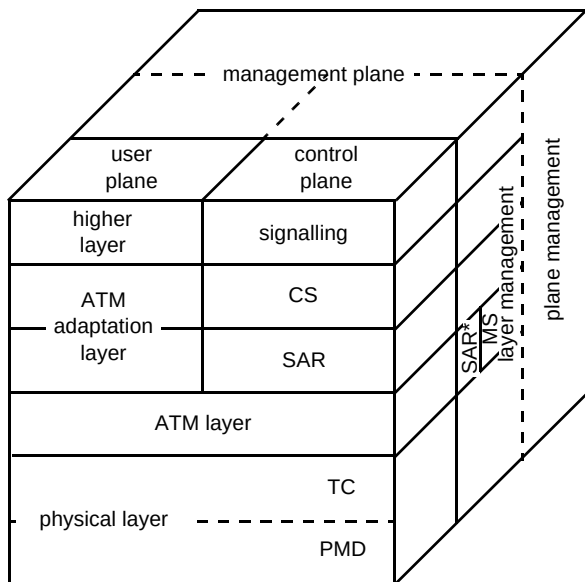


Fig 1 BUNI demonstrator architecture.



CS convergence sublayer
 SAR segmentation and reassembly
 SAR* segmentation and reassembly associated with metasignalling
 TC transmission convergence
 PMD physical medium dependent
 MS metasignalling

Fig 2 Cubic protocol reference model.

various specifications were to be implemented and to add a certain amount of implementation detail. The area that required the most development was the access signalling specification. At the time the BUNI specification was being produced little had been produced in the standards arena. Several other RACE projects had produced outline 'concept' specifications for the BUNI that were based on the ITU N-ISDN Q.931 specification. The BUNI demonstrator project used all the available information and after a considerable amount of effort produced a specification that fulfilled the requirements below:

- followed the emerging standards as far as practicable,
- contained enough detail to be implementable,
- had a high degree of functionality (service components, etc),
- interworked with N-ISDN.

The most notable aspects of the BUNI specification are that it included the concepts of metasignalling and service components. The metasignalling protocol allows the assigning, checking and removal of signalling channels. Within the BUNI demonstrator this provided little demonstrable functionality; it was concluded that metasignalling had little to offer a broadband network. This has recently, however, changed with the introduction of passive optical network (PON) technology into the local access network. Metasignalling can now be used to assign

access signalling channels, when needed, and hence save bandwidth on the PON. The other novel aspect of the BUNI demonstrator access signalling specification was the service component idea. This idea allows a broadband session to be broken down into a number of rudimentary components, e.g. speech, video, and fax. A user requiring a speech connection would send a 'SETUP' message to the network with one service component. At a later stage, if the user wished to send a fax to the same called party, a second service component (fax) could be added to the original call. The concept of service components was further refined by assigning each component a mandatory, optional or potential flag. The initial call set-up could therefore contain a number of components, some of which are mandatory (the call fails if these components cannot be set up), some of which are optional (the user would like these components but the call still succeeds even if these components fail), and some of which are potential (the user may wish to set up these components later in the call and is checking the compatibility of the called terminal). An example of this type of arrangement would be a video call, the speech component would be marked mandatory, the video component optional, and possibly a fax component marked potential. This would result in a call where, if the speech component succeeded, the user would also like a video component, but, if network congestion or terminal incompatibility caused it to fail, the call would still go ahead with the speech element, and the user would like to check the called party's fax capability as a fax component may be added to the call later. This arrangement allowed great flexibility for access signalling and service development but, as yet, has not been progressed to any extent within standards.

Other areas that were partially addressed within the BUNI specification were the management planes (layer and plane). It was found that the individual layers of the specification were fine, but, to allow a fully flexible terminal to be developed, a large degree of co-operation between the various layers of the specification was required. The type of actions that management had to co-ordinate were the spawning of new layer 3 access signalling entities for each new call, the binding of these access signalling entities to the higher and lower protocol layers, the creation and binding of user plane ATM adaptation layers to the higher layer application, etc. The BUNI specification included guidance on these issues.

The net result of the BUNI specification work was a non-formal document that was approximately 200 pages long and available as a single floppy disk.

So far this paper has only covered access signalling and its user and network access signalling entities. On the network side the network access signalling entity was supported via call and bearer control, neither of these two elements being to any international standards. These three

entities formed part of the network control architecture that also included the N-ISDN gateway and the configuration processors of the two ATM switches. The physical layout of the network control architecture had more to do with the split of work between the various companies involved than any technical requirements. All the parts that constituted the network control architecture were physically interconnected via an Ethernet LAN transporting BUNI command and control message set (BCCMS) messages — these messages were developed within the project.

2.4 Implementation

The implementation of the various signalling components were all hand coded, the user side access signalling being implemented by several companies, some within the BUNI demonstrator project and some from other RACE projects that brought their terminals to the BUNI demonstrator for testing purposes. The fact that a non-formal specification was being used, numerous companies all over Europe (not all English speaking) were involved, and hand coding was being employed, resulted in lengthy integration.

2.5 Lessons learnt

The process of developing a specification for the BUNI and the subsequent implementation of that specification by several companies coupled with the network control architecture created a number of learning opportunities:

- the development of a specification is a long, time-consuming process — it is much better to use standards wherever possible and customise them for the specific application (in most cases, for prototypes/demonstrations, international standards will be too complex, but it is easier to cut things out rather than put them in),
- once an agreed specification has been produced a rigorous change control procedure is required — this is very important when numerous companies throughout Europe are implementing the specification,
- if at all possible go for a formal specification (SDT was chosen for TRIBUNE, the BUNI demonstrator successor (see sections 3.4 and 4)) — the use of a formal language cuts down the ambiguities introduced in a plain English specification,
- implementation errors were only found when integration between two different implementations was tried — an implementation always works to the implementor's test harness,
- performance needs to be designed in not retrofitted, e.g. within the BUNI demonstrator the equipment was integrated and then the call set-up time measured and found to be rather slow — it is better to design in performance than try to improve an existing situation,
- the protocols on the control LAN used in the BUNI demonstrator were over complex — keep interfaces simple for performance, reliability and ease of implementation.

3. The ADP1 signalling demonstrator

The ADP1 signalling demonstrator was an internal BT project that built on the work done within the advanced broadband IN demonstrator (ABIND) project. The ABIND project had explored access signalling, intelligent networks (INs) and distributed processing; the ADP1 project added a videoconferencing application and a real transport layer that comprised ATM switching and PON local access technology. The final videoconferencing demonstration running over ATM was completed in early 1996 and was exhibited for Innovation '97 at BT Laboratories and the UK Alliance Symposium.

3.1 Demonstration environment

A videoconferencing application was developed for the ADP1 signalling demonstrator. The application included a number of user clients and a centralised service provider element. The dynamics of establishing a videoconference was that a user would send a videoconference request to the service provider; it was then the responsibility of the service provider to invite and establish connections to the other invited participants, i.e. the service provider provided a co-ordination and maintenance function. The service provider also performed the video/audio mixing function. The user terminals were PC based and included a special video card that allowed the user presentation control of up to four images (participants). The control was similar to the standard Microsoft windowing system, i.e. you could position, size, minimise, maximise any of the four windows. The fully functional PC-based conferencing terminals were augmented by two pseudo-terminals that were comprised of VTRs.

The network that supported the videoconferencing application included all the futuristic elements that BT foresees being in the network in the next 5 to 10 years, i.e. PON local access technology, ATM switching, network control running on a distributed computing platform, personal numbering, etc. Figure 3 shows the overall design.

3.2 Signalling

The basic concept behind the ADP1 signalling demonstrator was that all connections were controlled (established/released) via the access signalling protocols.

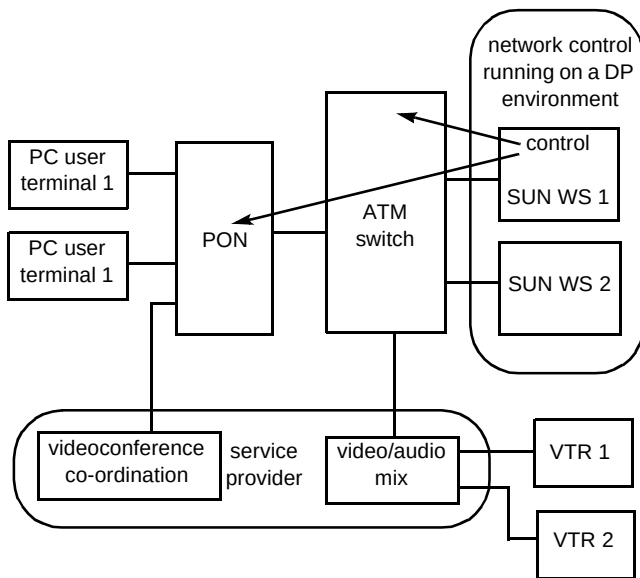


Fig 3 ADP1 signalling demonstrator architecture.

These access signalling protocols controlled two distinct types of connection — the low-speed control connection and higher speed video/audio connection. The low-speed connections were established between the user and service provider and carried a bespoke message set that requested the establishment of a videoconference session, invited users to a session, allowed users to leave a session, etc. The higher speed video/audio connections carried the users encoded audio/video in the user-to-service-provider direction and, in the reverse direction, the connection carried an audio mixed signal and a quadrature-encoded image of the videoconference participants. The quadrature image was used by the special video card to display the images of the participants in the separate user-controlled windows.

The network control elements included the network access signalling termination, call control, bearer control and a rudimentary IN layer. The IN layer provided number translation, personal numbering and user authentication type services. To allow personal numbering to function, the user has to register their location with the IN. This was achieved by extending the access signalling message set to include a non-call-related message based on the N-ISDN FACILITY message. The scenario was that before users could use any services they would first send a FACILITY register message to the network. This message would contain user identification and password information and at which location (line ID) they wish to be registered. This message would filter through the network access signalling and call control software to the service control element of the IN function. The user's information would be authenticated and, if correct, would be used to update the information held by the IN database. Subsequent requests to establish connection via the access signalling protocols would include the personal number of the called and calling

parties and the PIN of the calling party. With this information, call control would firstly request user authentication from service control and then number translation. Once call control knew the line IDs of the called and calling parties it would request bearer control to establish the connection. Bearer control was responsible for the allocation of virtual path/circuit identifiers (VPIs/VCIs) and the requests to the ATM switch and PON to physically set up the connection.

As can be seen from this environment, the access signalling message set has to be extended to include the FACILITY type register message and also to include additional information in the SETUP message for personal number and PIN information. In practice, the implementation of access signalling was rather weak and included only the message set, not the correct information elements within the message set nor the correct dynamics, i.e. no retransmissions nor timers implemented (see FINDS, in section 5, for a more standards-oriented implementation).

The only part of the jigsaw not discussed so far is the distributed computing element. All the network control software ran on a BT-designed distributed computing platform. The DP system was advanced in that it included trading and process creation features but not in a standardised way. The basic concept was that, when a message was received, a request would be made to the trader for the identification of a service resource manager that could create the required process to handle the message. Once the identity of the relevant service resource manager was known, it was requested to create the required process. This was repeated several times as a message filtered its way through the system.

3.3 Implementation

The BUNI demonstrator project illustrated that integration can be very difficult and time consuming if non-formal specifications are used. The decision was therefore taken within the ABIND and the ADP1 projects that the SDT tool-set from Telelogic should be used. This tool-set is based on the specification and design language (SDL) and includes a graphical input notation, simulation and auto-code generation. All of these elements were used on the development of the network code.

3.4 Lessons learnt

The development and integration of a large amount of software and hardware into a realistic demonstration that comprised most of the elements of a final system provided a number of important learning opportunities.

- SDT tool-set

It was found that the Telelogic SDT tool-set was good at developing a system in a very short time-scale. The basic inputting of the SDLs and subsequent simulations

was relatively straightforward. The final stage of auto-code generation for a particular target machine required a fair amount of 'black magic'. However, once the development environment had been established, the specification-to-code cycle was extremely short. To some extent this type of system requires managers to have 'nerves of steel' — there is no code until very near the end of the project, but when it appears it works nearly first time!

- **Software integration**

Integration of the software element that had been developed with the Telelogic SDT tool using SDL and automatic code generation was fairly easy. Few errors were found because the system had been fully simulated prior to integration. The problems experienced could be categorised into two types — logical and timing. The logical problems were easily found and fixes were relatively straightforward, i.e. go through the cycle of updating the graphical SDLs, simulation and re-autocode generate — a cycle that took less than 30 minutes. The timing-type problems were more difficult as the SDT tool-set did not have good debug facilities to capture this type of problem.

- **Performance**

Once again the performance was not designed into the system and hence call set-up times, etc, were rather slow. The requirement to investigate distributed computing necessitated the 'suck it and see' approach.

- **Appropriate technology**

The trial of an early distributed processing system was a good learning experience but, as it was not standards-compliant, the lessons learnt have to be carefully used. What is obvious is that any extra layer of complexity is going to slow down the system and plugging extra processors in does not always improve the performance — the software and the task being undertaken have to be appropriate to the technology.

4. The TRIBUNE testbed

The TRIBUNE project was an element of the RACE II programme. The aim of the programme, part funded by the European Commission, was to set in place a common infrastructure for advanced communications services, in particular the broadband integrated services digital network (B-ISDN), within Europe.

TRIBUNE stands for 'testing, ratification and interoperability of the broadband user/network interface'. This expression neatly encapsulated the essential project goals. There are two components to this project — firstly the broadband user/network interface (which is derived from BUNI) was identified as the key focus of concern for the project, and the second component covered what was to be

done to this interface, i.e. to prove that the interface was correct and complete by ratifying its definition, by showing that it could interwork to other services and networks, and finally by showing how this interface could be thoroughly tested. In order to verify the user/network interface in this way, the project designed and constructed a testbed located in the R&D laboratories of the Dutch PTT (KPN), near the Hague. The essential project goals were as listed below:

- to define and implement a specification for the broadband user/network interface,
- to provide feedback to the standards bodies on the base recommendations,
- to construct a thoroughly proven testbed, available to third party users,
- to demonstrate the interworking between TRIBUNE and other network types,
- to develop and utilise the necessary test methods and tools required by such an interface.

The TRIBUNE project was managed by a consortium of collaborating organisations comprising some 12 partners, from 8 European countries. The project commenced in 1992 and ran till the end of 1995.

4.1 Background

The TRIBUNE project came about as the natural successor to the RACE I BUNI demonstrator project (see section 2). TRIBUNE built on the successes of the BUNI demonstrator project by using the now available text from the ITU-T on the user/network interface, by designing robustness and performance into the network side to provide a stable and reliable testbed, through the use of sophisticated proprietary tools and the development of a class of purpose-built test tools, and by proving a wider range of interconnectivity types.

4.2 Testbed environment

In this section the design and usage of the TRIBUNE testbed by the project is described. The decision by the project consortium to design, construct and operate a testbed was taken early on. The nature of the testbed was determined essentially by the range of services that were targeted by the project design team. The types of services envisaged early on included:

- TV and HDTV services — to demonstrate the high bandwidths available through B-ISDN,
- multimedia services — to demonstrate the flexible transport and switching capabilities of asynchronous transfer mode (ATM),
- PC applications — to demonstrate the use of the broadband application programming interface (API),

- testers — to demonstrate a number of important testing features of the BUNI,
- ISDN services — to demonstrate the interworking between broadband and narrowband ISDN networks,
- LANs — to demonstrate the interconnection between standard office LAN equipment and the B-ISDN,
- external broadband services — to demonstrate the interoperability between separate B-ISDN testbeds.

TRIBUNE was essentially a pragmatic project; having developed the specification of the BUNI (see section 4.3) it was considered important to demonstrate that such a paper definition would work in practice. In the opposite direction, practical feedback into the BUNI specification from the testbed was anticipated. A number of objectives were set for such a testbed, the most important of which are listed below.

- Testbed set-up

The testbed should be quick to design. At the start of the project the expected duration was three years. To progress from initial project goals to a complete integrated testbed with public demonstrations within such a period, all within the context of an international, collaborative organisation, was a significant challenge.

- Reference model

The testbed was to serve as a reference model. From the beginning it was required to use the TRIBUNE testbed to prove a range of interworking scenarios. In addition, it was hoped to make the testbed available to as many user projects as possible. By widening the audience for the TRIBUNE BUNI specification, a greater degree of feedback would be obtained from implementors.

- Third party facility

This last aspect of the TRIBUNE testbed, its use as a third party testing facility, was a double-edged sword. Together with the advantages gained by such an approach, there were also constraints placed on the testbed. The first of these concerns robustness. In order to support a wide selection of equipment types being connected to the TRIBUNE testbed, there had to be an assurance that the network side could run unattended in a continuous and reliable fashion. It was also considered important that the testbed provide a reliable service to the 'users', and one which was within the expected performance margins agreed for the project.

The design of the complete TRIBUNE testbed is shown in Fig. 4. There were four essential parts to the testbed.

- The switching 'fabric'

Within TRIBUNE there were two separate switches. One of these was an SDH cross-connect, used for the distribution of TV and HDTV signals. The other was the ATM VCI/VPI switch; this represented the core of the network switching and transmission function.

- The testbed control network

This collectively represented the central B-ISDN exchange intelligence. At the centre of the control network was the TRIBUNE control server (TCS). This element was responsible for the control of calls across the TRIBUNE testbed, co-ordinating the actions of the other controllers as required in the establishment, modification and release of all the calls through TRIBUNE.

- Gateways

These extended the range of TRIBUNE by providing access to other network types. For example, one of the gateways had connections to the Dutch public N-ISDN; another one was connected to the European-wide ATM field trial network which was then undergoing test.

- The users

This term refers to the complete collection of user terminals and CPEs that connect to TRIBUNE over the published BUNI.

4.3 Signalling aspects

The TRIBUNE BUNI specification [1] was derived primarily from ITU-T Recommendations, approved documents where possible and draft recommendations where no fully agreed standard was available. At the time when the original specification was being compiled during late 1992 and early 1993, the ITU-T Recommendations were incomplete in a number of areas, particularly the ATM adaptation layer and access signalling. In these cases, additional information was taken from relevant ETSI material and ATM Forum specifications.

Considerable effort was made in compiling the TRIBUNE specification to make it complete, consistent and implementable. The majority of areas left 'for further study' in the standards documentation were resolved by the project.

A major objective in developing the TRIBUNE specification was that it should be implementable. This

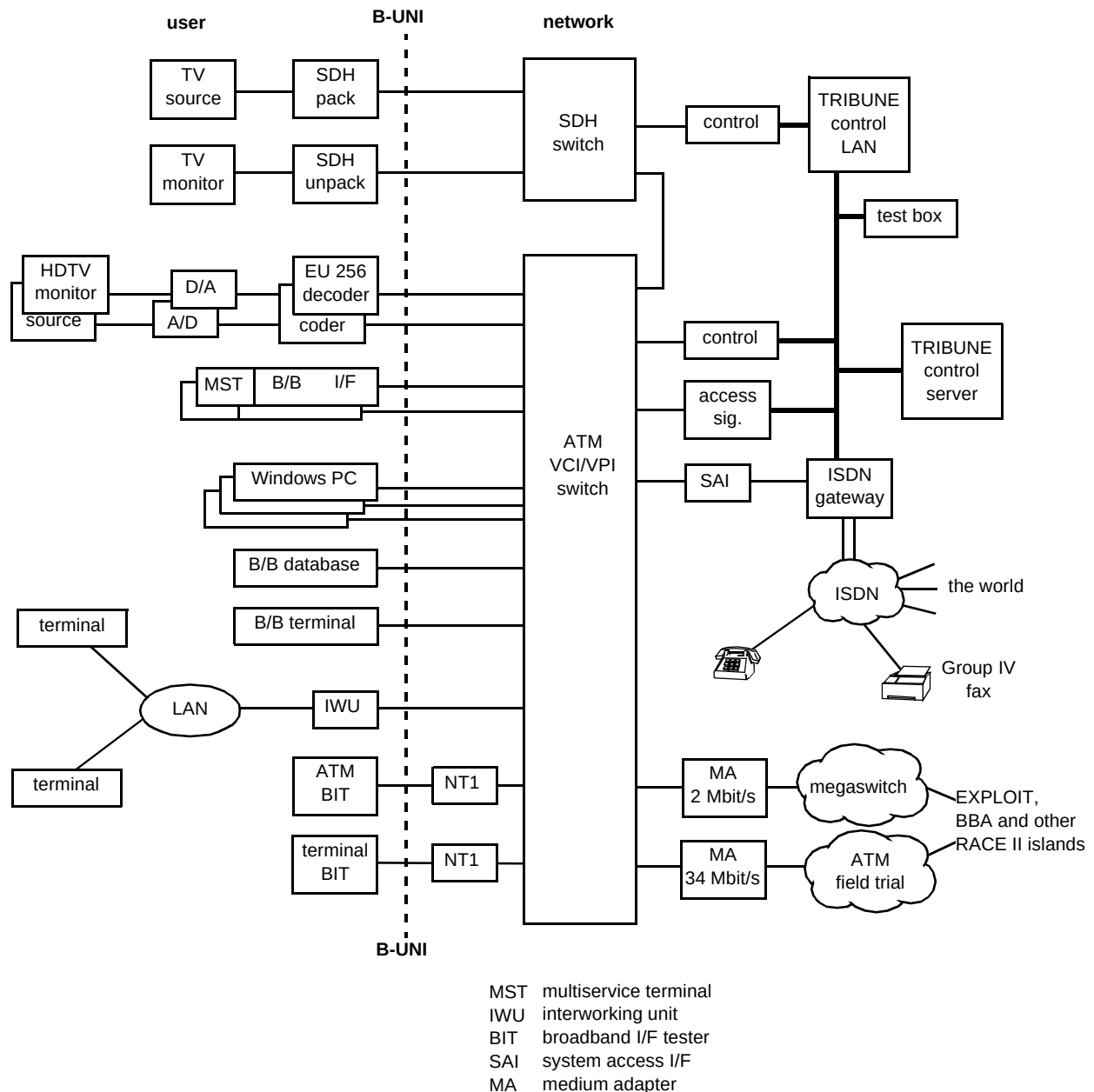


Fig 4 The TRIBUNE testbed design.

required that the text be consistent, but also that it be understood as easily as possible. Information was added to several parts of the document to illustrate the dynamic characteristics of its operation. A key aspect of this was the inclusion of overview message flow diagrams which were included to clearly explain the operation of the protocol. These were easily interpreted manually, giving an unambiguous, readable description of the protocol.

During the implementation of the specification, it was found that the text taken from ITU-T Recommendations could in many cases be clarified to aid correct implementation. The Recommendations were open to a degree of interpretation that was not suitable for a project such as TRIBUNE. A number of standards contributions

were made, and modifications made to the TRIBUNE specification to clearly specify what was required from implementations.

An essential element of the TRIBUNE BUNI specification was the use of formal methods in the upper layers of the associated protocol stack (see Fig 2 for the protocol reference model). Two separate SDL tools were used to specify the access signalling and ATM adaptation layers of the BUNI (on both sides of the interface). The use of SDL provided a much more rigorous (i.e. less ambiguous) definition of the user/network interface than was possible in the text-based version used for the BUNI demonstrator project (see section 2). The tools also provided automatic code generation as well as test case

specification. This increased automation of the higher level functions of the UNI resulted in a considerable saving in integration (and later in further adoption of the UNI to the ATM Forum definition of the interface).

4.4 *The lessons learnt*

The TRIBUNE project followed on, both in terms of its scope and in terms of the development teams involved, from the RACE I BUNI demonstrator project. In this way it benefited in a number of ways.

- Use of SDLs

The use of SDLs (and the tool-sets) was critically important. Although more time was spent 'up front' in the specification phase of the interface, this was more than compensated for by the shorter development time and the much shorter integration time than had been experienced by 'hand-coding' methods.

- Call processing times

The response time of the BUNI demonstrator network to new call attempts was poor (> 5 to 6 seconds to establish a point-to-point call). In TRIBUNE there was a requirement to return the equivalent of dial tone to an originating call attempt in less than one second. This target was comfortably met by local code optimisation, both by exploiting O/S features where appropriate, and by reusing the access signalling message set (defined at the UNI) over the service access point interface on top of the access signalling layer. In particular, this gave the network call control functions (CCFs) full access to whatever information it needed from that which had passed over the UNI. This design choice resulted in a number of benefits later on in the project.

- Integration of N-ISDN access

This was never properly achieved in the BUNI demonstrator project. In TRIBUNE, full access to a range of narrowband services was accomplished through a gateway device. The design of this unit was simplified by making the network CCF aware of the access signalling message content. Since these messages were themselves derived from their narrowband counterparts, it was a straightforward task to map between the two message sets in the N-ISDN gateway.

- ATM Forum interface

The project was extended to a fourth year in order to provide the testbed with an ATM Forum interface (V3.0). The use of SDLs in both the user and network sides of the UNI, and the extension of the access signalling UNI message set-up into the network CCF meant that it was possible to enhance the TRIBUNE testbed in a short space of time so that it could support

both the ITU and ATM Forum definitions of the broadband UNI (and indeed demonstrate interworking between the two interfaces).

- Proxy signalling

Providing the UNI access signalling messages into the network CCF meant that it was a simple matter to support calls between two terminals, only one of which had an access signalling capability. This form of proxy signalling (which was referred to as a $\frac{1}{2}$ -call) was effected by offering a simple human/machine interface (HMI) to the network support personnel. This HMI offered the requisite information that a normal, signalling, terminal would provide on top of the access signalling interface to the CCF. In effect, the CCF was almost unaware of the differences between a conventional call and the $\frac{1}{2}$ -call.

5. The FINDS platform

5.1 *Background*

The FINDS (fast Internet demonstrator and studies) project was a continuation of the ADP1 project described in section 3 and as such inherited an evolving broadband network demonstrator. Completed in early 1997, the FINDS demonstrator provided the capability of offering a wide range of broadband service demonstrations including one way, interactive and two-way services. The demonstration covered aspects of a broadband network including core switching, network/service management, access, IN and signalling, and advanced services.

The aim of FINDS was to enhance and refine the ADP1 demonstrator in the following areas:

- upgrade the IN/signalling network in the areas of performance and demonstrability,
- provide standardised access signalling (Q.2931) over a 25-Mbit/s BUNI interface,
- provide IN services based on the registration and mobility concepts,
- provide an upgraded form of the ADP1 video-conferencing application,
- demonstration of fast Internet access.

5.2 *Demonstrator environment*

IN services

The FINDS demonstrator was able to offer a number of IN services. The two services that were implemented were a personal numbering (PN) service and also a name calling service. The former was based on a traditional PN service. The latter gave subscribed users the ability to make a call to

another customer by using an alphanumeric name that was customised for the calling user, such as 'office', 'home', 'mum', etc.

Signalling demonstrability

One of the inherent problems of a demonstrator which focuses on signalling is the ability to display the actual signalling that is taking place. Viewers tend to concentrate on the application and take the signalling for granted. This was a particular problem that was found with the ADP1 project. For the FINDS project a graphical user interface (GUI) was developed which was a graphical representation of the signalling modules. This displayed the message flow between the modules, either in real-time or in a deferred message mode.

The interface also provided information on the logical state of each module, the data held within the IN databases, and details of the virtual paths and channels being used on the ATM switch at any given time.

Applications

In addition to connection-oriented calls, applications may also make connectionless calls across the FINDS network. The ATM switch is configured for LAN emulation and thus it is possible to communicate between terminals using IP signalling. This is particularly useful for applications that need to send or receive information regardless of time constraints, an example being a registration command between two terminals.

For initial demonstration purposes, a videoconferencing application was developed for the FINDS network. This application made use of both connection-oriented and connectionless communication. Each terminal had an IP address for connectionless communication and a terminal directory number for connection-oriented calls. The videoconference demonstration consisted of PC terminals at which the service users reside, plus a service bureau terminal from which the videoconference service was driven. At the user's PC terminal, the videoconferencing application was operated using a WWW browser. Each user would log in to the browser, the log-in being sent and confirmed via a connectionless communication with the service bureau. The service bureau retained a database of who was logged into the service so that conferences could be requested between people logged on to the service. When a videoconference session was initiated, the service bureau was informed by the conference requester, again via a connectionless communication. The service bureau would then request connection-oriented paths between itself and the participating terminals. It was then the job of the service bureau to receive, mix and return the video via the connection-oriented channels. Figure 5 shows the architecture of the demonstrator.

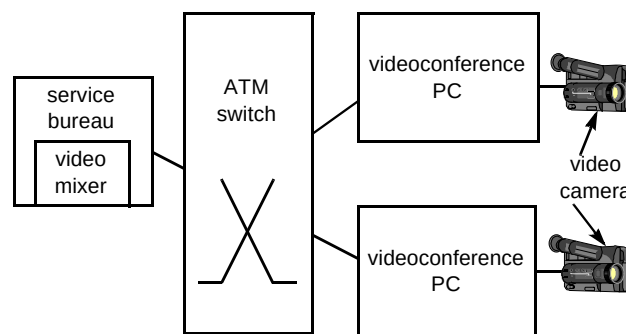


Fig 5 FINDS videoconferencing demonstrator.

5.3 Signalling

Figure 6 shows the modules that make up the network control signalling software. Each module runs as a separate Unix process and communicates via socket interfaces. The modules have the following functions:

- Call control module

The call control module links the two ends of a call, the calling and called parties. It achieves this by co-ordinating the access signalling messages to/from each user with the appropriate physical end-to-end connection that is needed to transport the information for the requested service — be it voice, data, images or whatever.

- Service control module

The service control module adds intelligence on to call control. Call control can typically establish a basic call to a directory number when requested by the calling party, and service control is used for the IN-type calls such as personal numbering and name calling.

- Bearer control module

The bearer control module is sometimes referred to as connection control. It manages the physical connection established between the two ends of a call. Bearer control receives requests from call control during the lifetime of a call and interfaces to the ATM switch to establish or remove the relevant circuits.

- Access signalling module

The access signalling module manages the Q.2931 protocol that operates between the user and the network. This gives the user (terminal equipment) the ability to access and control the wealth of broadband/multimedia services and capabilities that are inherent within the B-ISDN network.

5.4 Access signalling

The access signalling software for both the user and network sides is derived from SDL specifications. The base

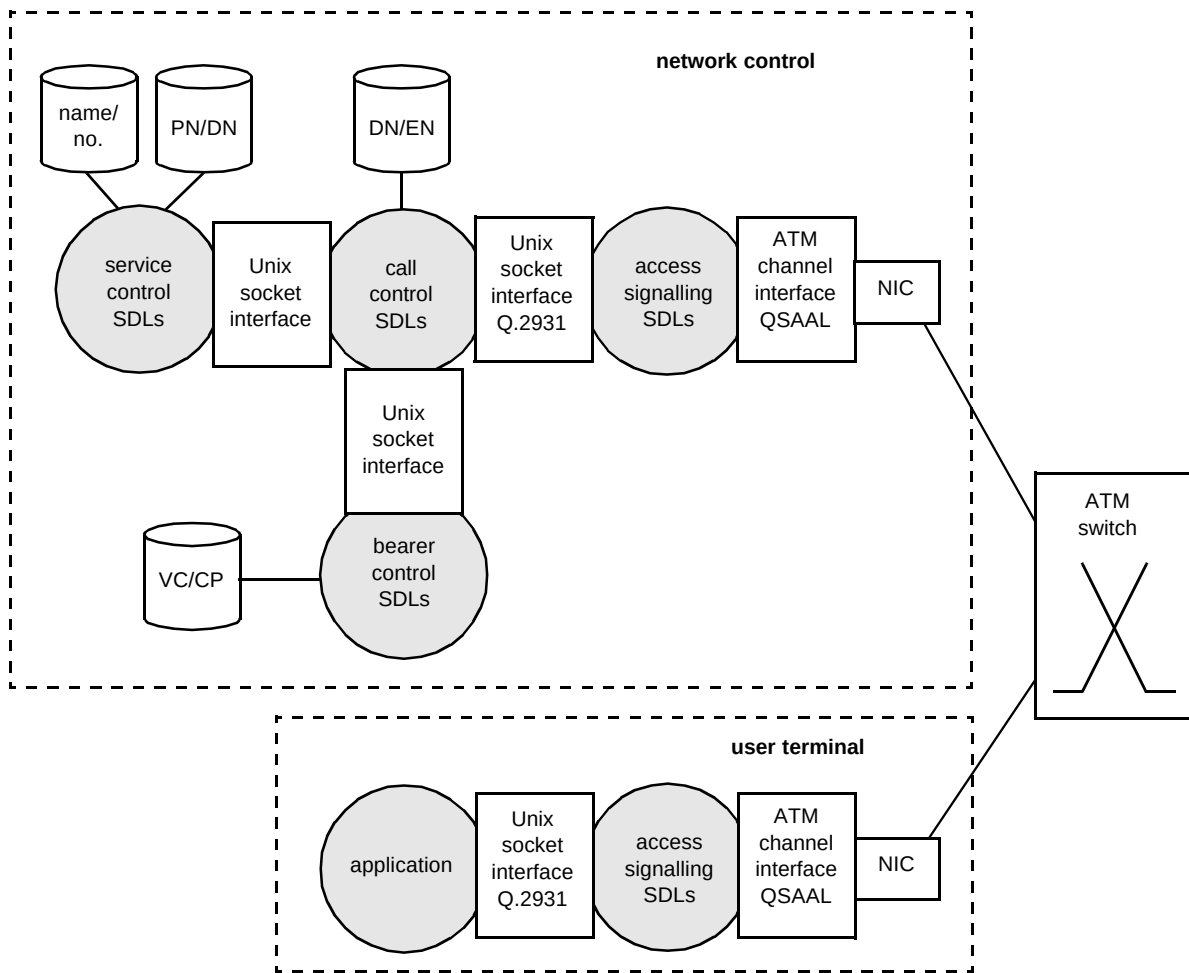


Fig 6 FINDS architecture diagram.

SDLs used in the FINDS project were taken from the RACE II TRIBUNE project [1] (described in section 4). Two packages of SDLs from the TRIBUNE project were used, the Q.2931 and signalling ATM adaptation layer (QSAAL) systems. Editing of the SDLs was required to make them meet the needs of the FINDS project both architecturally and in terms of the messages and information elements supported.

With the SDLs in the required state, and simulations of the SDL behaviour produced to verify their operation, conversion of the SDL to C code was performed, the derived C code being used to produce the final executable access signalling module. In conjunction with the SDL derived code, 'hand-produced' C code was written to provide an interface into the SDL-produced behaviour. Figures 7 and 8 illustrate the architecture of the access signalling module, both for the network and user sides respectively, and show the added interface functionality.

The structures for user and network side are very much the same. In the network side, a QSAAL/CRV (call reference value) mux instance is required for each terminal that the network supports. The QSAAL process supports the

low-level QSAAL connection between user and network. The CRV mux is used to initiate an access signalling (AS) process for each active call between a user and the network. Initially there are no calls and no AS processes exist. When a call is made either from or to a terminal, the CRV mux instantiates an AS process, while the CRV mux records which AS processes are driving which calls via the CRV. The AS process is equivalent to Q.2931. In the network the upper equipment number (EN) mux and lower EN mux are used to send the correct signalling messages to the correct QSAAL/CRV mux instance. The EN muxes use the equipment number of the terminal at the user's side to send the signalling messages to the network processes tracking the calls to that terminal.

Outside the SDL derived code area, 'hand-written' coder/decoder functions are used to convert the SDL formatted messages into octet strings that can be passed to other software modules. The coding of the messages comply with the Q.2931 standard [2]. The coded octet strings are, at the QSAAL end, sent to and received from the ATM network interface card both at the user's terminal and the network control terminal. At the Q.2931 end the coded octet strings are sent to and received from socket interfaces

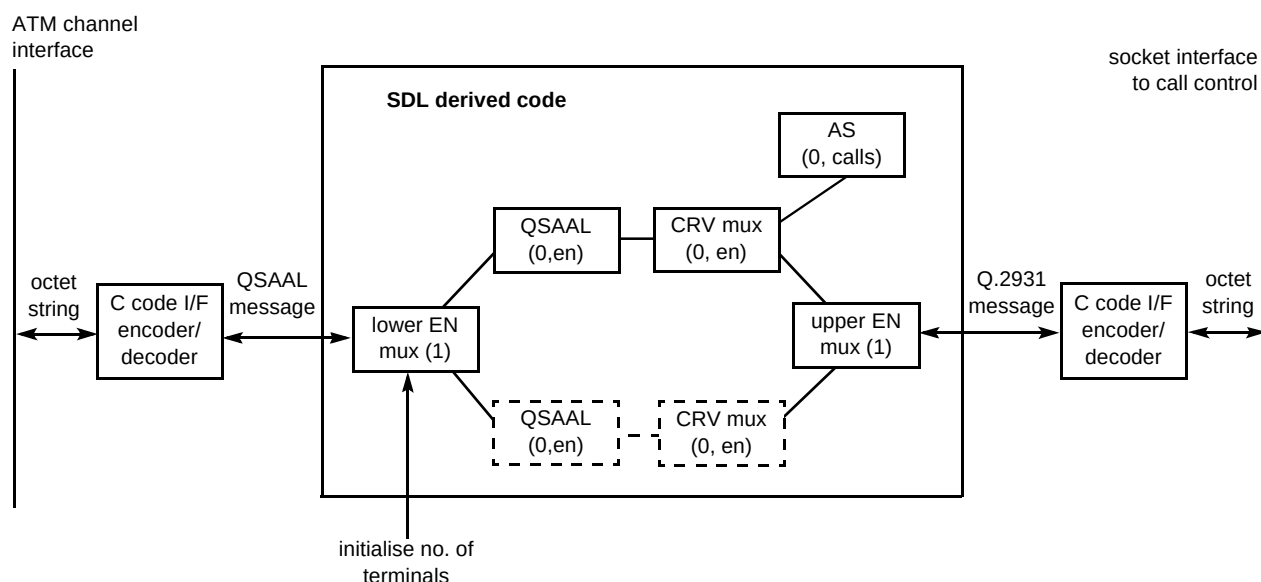


Fig 7 Network side access signalling components.

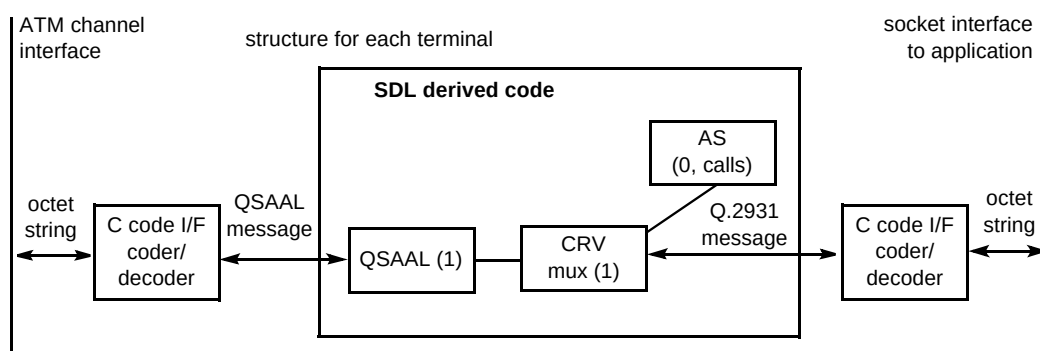


Fig 8 User side access signalling components.

to other modules of code (call control at the network side, the controlling application at the user side).

For the FINDS project a defined access signalling API [3] exists to aid application development. A defined FINDS API is required since the project only uses a subset of the full Q.2931 signalling messages, and within those messages only a subset of the Q.2931 information elements. Part of the API package is the FINDS Q.2931 coder/decoder mentioned above. As long as the access signalling messages are structured correctly in the resident application, the coder/decoder will ensure that the octet string passed to the access signalling module is correct. In addition to defining the Q.2931 messages and information elements, the API describes the process for registering a service type and for requesting a CRV. Since the FINDS network has been designed so that multiple applications may reside on each user's terminal, some mechanism was needed for ensuring that access signalling messages are directed to the correct application. In order to do this, an application must register the service it provides (e.g. videoconference, 64-kbit/s speech) prior to any message exchange. All incoming call set-up messages arriving at a user-side access signalling

will have a defined service type. The access signalling uses the service type to determine which application should receive the set-up message. Additionally CRV registration must take place within the access signalling module — this ensures that none of the applications are using the same CRV value. An application wishing to initiate a call must apply to the access signalling module for a CRV prior to sending the call set-up message.

5.5 Implementation

As with the ABIND and the ADP1 projects discussed in section 3 the SDT tool-set from Telelogic was used to build the code for each of the software modules shown earlier in Fig 6. This was an obvious decision having experienced its advantages from the earlier projects as well as having the base access signalling SDLs.

Figure 6 shows the basic software modules that were built using the SDT tool-set, along with the Unix socket interfaces which were manually coded. The access signalling socket interfaces, also incorporated a message coder and decoder to convert the access signalling message set to/from the SDT text message format from/to the bit

message format, as specified by the Q.2931 standard. This gave the platform a standardised access signalling interface.

5.6 Lessons learnt

The evolution of this broadband network demonstrator reiterated a number of lessons previously learnt, as well as providing some additional learning opportunities.

- Call set-up times

The performance of the demonstrator was significantly improved over that of the ADP1 demonstrator. Call set-up times were reduced to an acceptable 2 seconds. This was primarily due to the vast reduction in the number of dynamic Unix processes that were previously being created during a call set-up.

- Third party layer

The performance was also improved by removing a third party distribution layer between the software modules and replacing it with an integrated socket interface. This reconfirms that an extra layer of complexity slows the system down.

- SDT tool-set

It was reiterated that using the Telelogic SDT tool-set was a good approach to developing a system in a short time-scale, by simulating and testing a set of SDLs before automatic code generation.

- Access signalling

One of the more interesting developments during the testing of the FINDS network was the performance of remote links to the access signalling, the requirement being to allow the videoconference application to be physically tested remotely from the FINDS network. This was achieved by configuring the access signalling module on one of the terminals to accept access signalling messages across an IP socket. The IP socket connection used the BT Laboratories' site backbone as the transport medium, this being connectionless with no guaranteed quality of service. This type of testing provided a useful gauge of the differences in performance and quality between terminal resident access signalling and proxy access signalling. What was noted in the process was that the reliability of the network between application and proxy access signalling is an important factor in the success of this method. Over the BT Laboratories' network the connection was not always reliable and on some occasions remarkably slow. Over a much shorter distance, such as a single LAN subnet, the proxy signalling connection exhibited similar connection and speed characteristics as a terminal-resident signalling connection.

5.7 Future plans

As mentioned previously, the FINDS network was initially developed with only the videoconference application to provide a demonstration of the potential services that may exist on the platform. The network was intended to provide a base for operating any broadband application, the only criteria placed on the application developer being adherence to the defined access signalling API, described above, for creating connection-oriented connections across the network. It is hoped that future services will be developed for evaluation using the FINDS network. One current development is the porting of the DAVIC broadcast TV demonstrator [4] on to the FINDS network. The DAVIC demonstrator uses a similar ATM network configuration to FINDS, but uses a different access signalling method to FINDS, proxy signalling being preferred to terminal-resident signalling. Porting the application on to the FINDS network requires an alteration to the existing interface to access signalling. In terms of the DAVIC 'S' layers, a FINDS 'S3' layer is required in order to interface between the current DAVIC 'S1' and 'S2' layers and the FINDS access signalling 'S4'.

Another current activity is a project to develop a Java terminal to run on top of the FINDS network. At its heart, the Java terminal consists of a harness to enable Java-produced access signalling messages to converse with the user terminal access signalling module (see Fig 9). At its lower, access signalling interface, the Java harness is based upon the FINDS access signalling API. The upper interface to the Java harness is JTAPI (Java Telephony API) compliant. JTAPI is a standardised Java interface for telephony-type Java applications. This JTAPI interface to

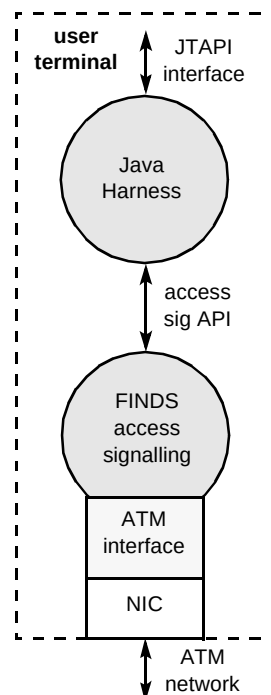


Fig 9 FINDS Java harness.

the Java harness allows any Java application, written to use the JTAPI interface, to operate over the FINDS network. Since Java is a fast developing language, and BT is keen to exploit its use, the existence of the JTAPI interface on the FINDS network will enhance its usefulness.

6. Conclusions

This paper has described four projects which vary considerably in their nature and scope, but which were all concerned in an essential way with the design, development and support of broadband signalling interfaces. A small number of common themes have arisen from these projects; these are captured below.

- Interface design

Whenever possible, resist the temptation to devise your own interfaces. They are almost always more difficult to design and maintain than would be first imagined. Use standards-based interfaces when available, or at least the best parts of such.

- Reuse of interfaces

As a corollary to the previous point, make every effort to reuse other interfaces within your project. In TRIBUNE the BUNI access signalling level interface was used in its entirety at the AS-SAPI (i.e. into call control within the network). This not only saved the effort of maintaining a separate interface definition but (paradoxically though it might seem) also contributed to performance optimisation.

- Standard interfaces

Keep interfaces as simple as possible. The FINDS project has shown the benefits of employing standard definitions such as IP sockets, and is now exploiting the portability features of JAVA interfaces.

- Formal protocol language

The definition of all but the simplest levels of protocol behaviour should use formal techniques based on SDL specifications and associated tools.

- Non-functional requirements

Non-functional requirements (e.g. performance criteria, reliability) should be designed in from the outset, not added afterwards.

References

- 1 RACE II TRIBUNE: 'TRIBUNE BUNI Specification', internal BT document (1995).
- 2 ITU-T Recommendation Q.2931: 'B-ISDN-DSS2 — user/network interface layer 3 specification for basic call/connection control', (1994).
- 3 King J R: 'Access signalling interface', internal BT document (1996).
- 4 Reece P W, Macey R D, Galbraith R I and Gentry P: 'Plan, design and specification for the implementation of separate STU and NIU functionality on the DAVIC session control demonstrator', internal BT document (1996).



Chris Shephard read Theoretical Physics at the Universities of East Anglia and then Birmingham. He joined BT in 1980 and worked for a number of years on the development of the System X switching systems. On the basis of this experience he accepted a job with Siemens in Florida within a large project designed to adapt their range of digital public switches (EWSD) to the US market and regulatory requirements. He returned to BT after two years to work on local network access switches and later on broadband ISDN systems (during which time he was the BT project manager for

TRIBUNE). He is now part of a development group established to exploit the capabilities of network computers.



Peter Hovell graduated from the Royal Military College of Science in 1978 with a BSc degree in Electrical Engineering. After a number of years working for the civil aviation authority on the development of radar processing equipment, he joined BT. Initially he worked on the system specification and design of novel local access mechanisms before becoming involved with two RACE projects dealing with broadband ATM demonstrators. During the last few years he has been associated with a number of projects constructing broadband demonstrators at BT Laboratories and is currently leading an

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John King graduated in 1989 with a BSc(Hons) in Electronics and Microcomputer Systems from the University of Dundee. He joined the signalling and protocols unit at BT Laboratories where he was involved in investigating methods for formally specifying and testing signalling protocols.

Latterly he has moved to the Intelligence and Internet Engineering unit working in the broadband and Internet arena. He has an MSc in Telecommunications for BT from the University of London.